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Predicting attitudes toward cryptocurrencies and stocks: The divergent roles of narcissism, intelligence and financial literacy

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ABSTRACT

Relatively narcissistic people are attracted to cryptocurrencies, though it is unclear whether they are differentially attracted to cryptocurrencies over other investments. Furthermore, theoretically, only narcissistic admiration, rather than narcissistic rivalry, would be expected to associate with attitudes toward cryptocurrencies. Intelligence and financial literacy are also proposed individual difference predictors of attitudes toward investments. Consequently, we administered measures of narcissistic admiration and rivalry, a financial literacy test, and a battery of intelligence tests to a sample of young adults ($N = 372$). Based on a structural equation model, narcissistic admiration and narcissistic rivalry differentially predicted attitudes toward cryptocurrencies (admiration, positively; rivalry, negatively), but both failed to associate significantly with attitudes toward stocks. Furthermore, financial literacy was a unique, positive predictor of attitudes toward stocks, whereas intelligence was a unique, negative predictor of attitudes toward cryptocurrencies. Our findings support the notion that narcissism is differentially associated with attitudes toward cryptocurrencies, though only narcissistic admiration (positively), consistent with the hypersensitivities to reward theory. Finally, higher levels of intelligence, controlling for financial literacy and narcissism, associated negatively with attitudes toward cryptocurrencies, perhaps due to the influence of scepticism.

1. Introduction

Cryptocurrencies are an increasingly popular investment (Afzal & Asif, 2019). As cryptocurrencies are highly volatile (Shen et al., 2020), significant investment losses (and gains) can be realised. Additionally, young adults are disproportionately attracted to cryptocurrencies (e.g., Fujiki, 2021), however, they can usually least afford to lose their relatively few savings (Cribb et al., 2016). Consequently, understanding the psychological predictors of young adult attitudes toward cryptocurrencies may be considered important.

Recently, Martin et al. (2022) reported a positive correlation between narcissism and attitudes toward cryptocurrencies. However, Foster et al. (2011) found that narcissism correlated positively with the degree of risk in stock market selections. Because Martin et al. (2022) measured only attitudes toward cryptocurrencies, it is not known whether narcissists are differentially attracted to cryptocurrencies, in comparison to other risky asset classes. Another limitation associated with Martin et al. (2022) (and Foster et al., 2011) is that they measured narcissism in a unidimensional manner.

1.1. Narcissism dimensionality

Narcissism consists of three dimensions: (1) narcissistic admiration (a.k.a., agentic), (2) narcissistic rivalry (a.k.a., antagonistic), and (3) narcissistic neuroticism (Miller et al., 2021; Sedikides, 2021). Narcissistic admiration is defined by an overly inflated sense of self and abilities, an excessive sense of entitlement, and an intense need to be the centre of attention (Back et al., 2013). By contrast, narcissistic rivalry is defined by a self-defensive tendency that manifests in the devaluation of others and a desire to see others fail, which is motivated by the desire to feel superior to others (Back et al., 2013). Narcissistic admiration is considered the default/dominant expression of narcissism, whereas narcissistic rivalry is secondary and arises reactively when the ego is threatened (Grapsas et al., 2020). Together, narcissistic admiration and narcissistic rivalry form a higher-order dimension known as grandiose narcissism (Miller et al., 2021).

As grandiose narcissism is associated with a desire to be seen as superior, and as wealth confers a sense of prestige and power (Ahuvia, 2008), one may expect to observe a positive association between

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grandiose narcissism (admiration and rivalry) and attitudes toward investing (stocks and cryptocurrencies). Furthermore, one may hypothesize that grandiose narcissism would associate more substantially with cryptocurrencies than stocks, as cryptocurrencies are more volatile than stocks, i.e., greater opportunities for reward (Shen et al., 2020).

However, narcissistic admiration and rivalry predict criteria differentially (Back et al., 2013). For example, narcissistic admiration, but not rivalry, correlates positively with the overestimation of intelligence (Zajenkowski, Leniarska, et al., 2020). Additionally, narcissistic rivalry is associated negatively with trust and inclination to forgive, and positively with negative affect, whereas as narcissistic admiration is not (Back et al., 2013; Fatfouta et al., 2015). Theoretically, a positive association between narcissistic admiration and attitudes toward stocks/cryptocurrencies may be hypothesized, as narcissistic admiration is uniquely linked to hypersensitivities to reward and high approach motivation (Grapsas et al., 2020). Thus, a person high on narcissistic admiration may be expected to view cryptocurrency volatility as an attractive opportunity for rewards. By contrast, we hypothesized that narcissistic rivalry would associate negatively with attitudes toward stocks/cryptocurrencies, as narcissistic rivalry is theoretically and empirically linked with a fear of failure and cynicism (Lange et al., 2016; Miller et al., 2021). Thus, a person high on narcissistic rivalry may see stocks/cryptocurrencies as an opportunity to lose money, not gain.

1.2. Financial literacy and intelligence

Financial literacy represents people's knowledge and understanding of basic financial terms and concepts (Hastings et al., 2013). Correspondingly, financial literacy is associated positively with the likelihood of owning stocks (Kim et al., 2019; Thomas & Spataro, 2018). Thus, we hypothesized that there would be a positive association between financial literacy and attitudes toward stocks. In contrast to stocks, Zhao and Zhang (2021) found financial literacy associated negatively with cryptocurrency ownership. However, Fujiki (2021) reported a small, positive correlation with cryptocurrency ownership. Unfortunately, research in the area tends to measure financial literacy with five or fewer items, a psychometrically inadequate number (see Gignac & Ooi, 2022). Additionally, several factors may influence whether a person owns an investment or not, including funds surplus to one's expenses (i.e., a non-psychological characteristic). Consequently, it may prove more insightful to measure people's attitudes toward cryptocurrencies (and stocks), alongside a more psychometrically robust approach to financial literacy measurement.

In addition to financial literacy, intelligence has been postulated to be one of three primary predictors of investment awareness and participation (Cheng et al., 2018). Grinblatt et al. (2011) found that stock investors had significantly higher IQs ($d \approx 0.50$) compared to non-investors, even after controlling for income. However, it remains to be determined whether the positive effect between intelligence and stock market attitudes is a unique effect, or whether it is substantially accounted for by financial literacy, as intelligence and financial literacy share substantial variance (Gignac et al., 2023; Lin & Bates, 2022).

1.3. Summary and purpose

One study suggests that narcissism may be a positive correlate of attitudes toward cryptocurrencies (Martin et al., 2022). However, the issue has not yet been examined from a multidimensional (narcissism) perspective. Theoretically, it may be expected that narcissistic admiration would associate positively, and narcistic rivalry negatively, with attitudes toward cryptocurrencies, as narcissistic admiration is characterised by a high approach motivation (hope for success; Grapsas et al., 2020), whereas narcissistic rivalry is characterised by a fear of failure (Lange et al., 2016).

Additionally, both financial literacy and intelligence may be expected to associate positively with attitudes toward stocks (Grinblatt et al., 2011; Thomas & Spataro, 2018), though the expected direction of the association with cryptocurrencies is less clear (Fujiki, 2021; Zhao & Zhang, 2021). Consequently, we tested a structural equation model to estimate the unique and potentially differential effects of narcissism (admiration and rivalry), financial literacy and intelligence, as simultaneous predictors of attitudes toward cryptocurrencies and stocks.

2. Method

2.1. Sample

A total of 372 first-year (English as a first language) psychology students completed the testing for course credit at a large Australian university. Missing data were substituted for three cases (see Note 1 in supplementary materials). The sample was relatively young (M_{age} = 19.78 years, SD = 4.11), predominantly White (White = 68.3 %; Asian = 25.3 %; Black = 1.3 %; Hispanic = 0.8 %; Indigenous = 0.8 %; Other = 2.7 %), but relatively balanced for sex (57.5 % female). Median weekly income was $300 ($M$ = $349.60, SD = 306.00; $0 = 13.4 % of the sample). Power was calculated at 0.97 to detect a correlation of $|0.20|$ as significant ($p < .05$).

2.2. Measures

Attitudes toward cryptocurrencies (ATC) were measured with three questionnaires. First, consistent with Martin et al. (2022), the question "What do you feel about cryptocurrency?" was rated on three separate seven-point Likert scales: "unfavourable", "negative", "bad" (=1) to "favourable", "positive", "good" (=7). Additionally, the question. "If you were looking to invest, how likely are you to buy cryptocurrency?" was rated on three separate seven-point Likert scales: "unlikely", "definitely would not", "improbable" (=1) to "likely", "definitely would", "probable" (=7). Finally, an adapted four item self-reported measure (5-point Likert scale; 1 = '*strongly agree*'; 5 = '*strongly agree*') of attitudes toward stocks by Keller and Siegrist (2006) was used, where the word 'stock' was replaced with 'cryptocurrencies'. An example item included, "When I hear the word 'cryptocurrencies' the term 'possible loss' comes to mind immediately." After reverse scoring (three items), the item scores were summed and averaged (response scale: 0–5), with higher scores reflecting a greater willingness to invest in cryptocurrencies.

Attitudes toward stocks (ATS) were measured with the same three above questionnaires. However, with respect to the two questionnaires from Martin et al. (2022), the word "cryptocurrency" was replaced with "stocks". Additionally, the original (non-adapted) four items from Keller and Siegrist (2006) were used. In order to calculate participant composite scores for overall attitude toward stocks and overall attitude toward cryptocurrencies, the individual questionnaire composite scores were transformed into z-scores. Then, the three respective z-scores were averaged together to form two composite variables, ATS.z (ω = 0.84) and ATC.z (ω = 0.83). See Note 2 in supplementary materials for justifications on the combination of the attitude/intention measures.

Narcissism was measured with the Narcissistic Admiration and Rivalry Questionnaire (Back et al., 2013). Internal consistency reliability in this sample was good (NARQ-Admiration: ω = 0.84; NARQ-Rivalry ω = 0.76). Intelligence was measured with three tests. Speed of reasoning (*Gs/Gf*) was measured with the Baddeley 3-min Reasoning Test (Baddeley, 1968; this sample ω = 0.93). Visual-spatial ability (*Gv*) was measured with Form A and Form B (3.5-min time limit for each form) of the Paper Folding Test (Ekstrom et al., 1976; this sample ω = 0.85). Crystallised intelligence (*Gc*) was measured with the 50-item Single Word Comprehension Test (Warrington et al., 1998; this sample ω = 0.70).

Financial literacy was measured with the Fernandes et al. (2014) test and three newly created questions to help enhance test score reliability. The questions measured knowledge of inflation and the time value of

money, the benefits of diversification, and the nature of compound growth, for example (see Note 3 in supplementary materials for questions). This sample reliability was $\omega = 0.62$.

2.3. Procedure

Testing was completed remotely via Qualtrics (online). After providing informed consent (Australian University of Australia Human Research Ethics Committee, 2020/ET000269), participants completed the demographic questions, the Baddeley Reasoning Test, the Paper Folding Test, the Synonym Test, the financial literacy test, and the attitudes toward cryptocurrency and stocks questionnaires (order randomized). For a different project, the participants also completed a financial wellbeing, financial locus of control, money values, and monetary choice questionnaires. Total testing time was approximately 40 to 45 min. Data were collected between May 2, 2022, and August 5, 2022, a period in which the Bitcoin price (AUD) and the S&P/ASX 200 declined by 41 % and 13 %, respectively (at bottom).

2.4. Data analysis

All analyses were conducted with SPSS and Amos (versions 27) and Jamovi (version 2.2.5). Cognitive ability test sum scores were converted into percentage correct scores to facilitate interpretation. With respect to the latent variable model, model close-fit was evaluated via RMSEA and SRMR (≤ 0.08 acceptable close-fit), as well as CFI and TLI (≥ 0.950 acceptable close-fit). Maximum likelihood estimation and the confidence intervals and *p*-values were estimated via bias-corrected bootstrapping (2000 resamples). Given its substantial correlation with intelligence (Gignac et al., 2023), financial literacy test scores were specified as an indicator of general intelligence, but we also specified direct effects between financial literacy and the attitudes toward stocks/cryptocurrencies latent variables (see Fig. 1). Data and variable descriptions can be found here: https://osf.io/c6gq5/

3. Results

3.1. Descriptive statistics

Though a small number of outliers were identified for some variables (see Note 4 in supplementary materials), we used bootstrapping to estimate the confidence intervals and *p*-values for all analyses, as bootstrapping does not assume normality. A total of 16.9 % of the sample owned stocks ($n = 63$) and 11.6 % owned cryptocurrencies ($n = 43$). Additionally, if the participant was looking to invest money, 21 % and 53 % of the sample said they would be likely to invest in cryptocurrency and stocks, respectively (i.e., scored $\geq$5 on the relevant item; see Note 5 in supplementary materials). Thus, there was appreciable variability in the sample's ownership and attitudes toward stocks and cryptocurrencies.

The correlation between the attitudes toward cryptocurrencies composite and the attitudes toward stocks composite was $r = 0.33$, 95 % CI[0.22, 0.43], $p < .001$. An examination of the scatter plot (Jamovi flexplot; loess regression) revealed that the association was nonlinear (see Fig. 2). Furthermore, there appeared to be two clusters of people beyond an average positive attitude toward stocks: those with a positive attitude toward cryptocurrencies and those with a negative attitude toward cryptocurrencies.

3.2. Observed and latent variable correlations

The supplementary materials include the observed score correlations and descriptive statistics (see Table S1). As can be seen in Table 2, the NARQ-Adm and NARQ-Riv latent variable correlation was positive and appreciable ($r = 0.57$, $p < .001$), as expected. Additionally, the attitude toward stocks (ATS) and attitude toward cryptocurrencies (ATC) latent variable correlation was $r = 0.48$, 95 % CI [0.32, 0.58], $p < .001$ (see Table 2), suggesting convergence but also distinctiveness. The IQ*g* latent variable correlated positively with ATS, $r = 0.28$, 95 % CI [0.12, 0.45], *p*

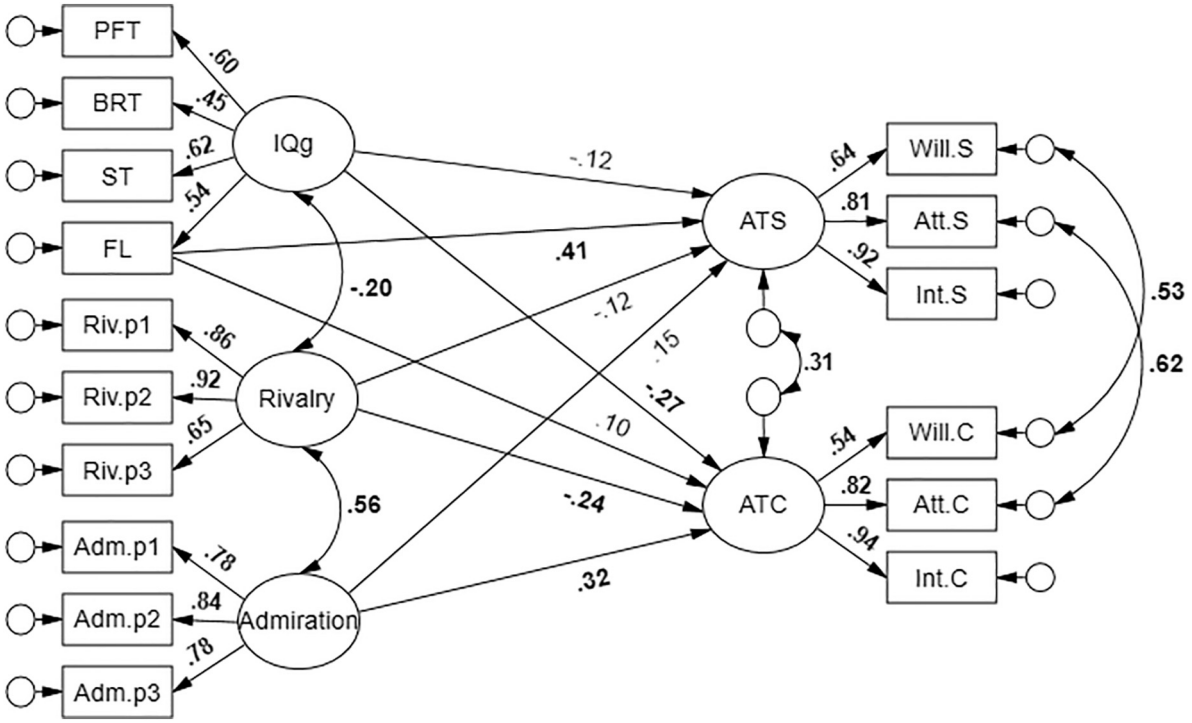


Fig. 1. SEM Depicting the Fully Standardized Associations
Note. $N = 372$; PFT = Paper Folding Test; BRT = Baddeley Reasoning Test; ST = Synonym Test; FL = financial literacy; Riv = narcissistic rivalry; Adm = narcissistic admiration; .p = parcel; IQg = general intelligence; ATS = attitude toward stocks; ATC = attitude toward cryptocurrencies; Will = willing to invest; Att = attitude toward; Int = intention to buy; .S = stocks; .C = cryptocurrencies; coefficients in bold were significant, $p < .05$.

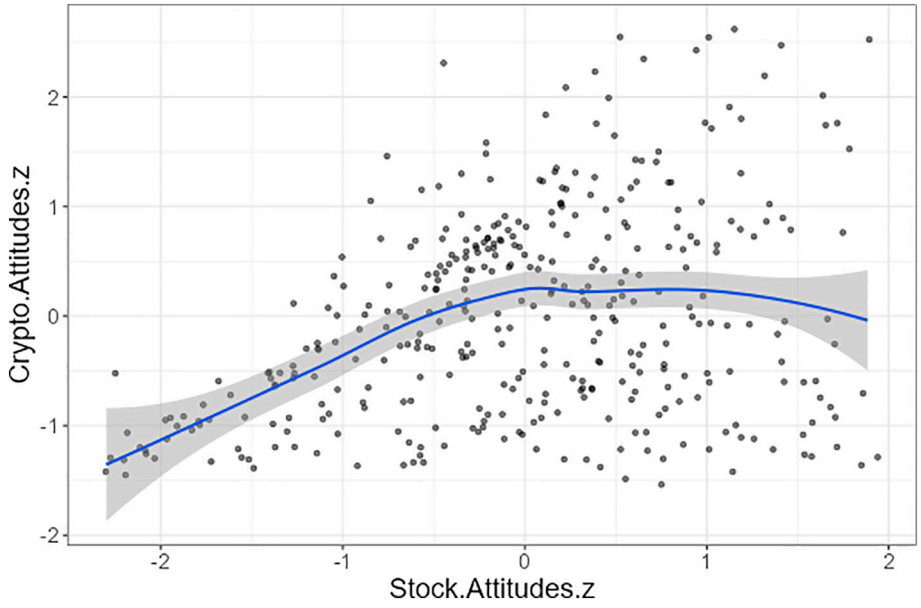


Fig. 2. Scatter Plot Depicting the Association between Attitudes Toward Stocks and Attitudes Toward Cryptocurrencies
Note. $N = 372$; line of best fit = loess; jitter (0.25 in X and Y); shaded area represents 95 % confidence bands.

Table 2
Latent variable correlations & reliabilities.

	1.	2.	3.	4.	5.	6.
1. NARQ-Adm	(0.84)					
2. NARQ-Riv	**0.57**	(0.86)				
3. ATS	0.06	−0.06	(0.84)			
4. ATC	**0.21**	0.00	**0.48**	(0.83)		
5. IQ*g*	−0.13	**−0.26**	**0.28**	**−0.16**	(0.64)	
6. Earnings	−0.06	−0.06	**0.14**	0.01	**0.05**	(–)

Note. $N = 372$; NARQ = Narcissistic Admiration and Rivalry Questionnaire; Adm = admiration; Riv = rivalry; ATS = attitudes toward stocks; ATC = attitudes toward cryptocurrencies; IQ*g* = general intelligence; earnings = weekly earnings; coefficient omega on main diagonal in parentheses; coefficients in bold were significant, $p < .05$; see Table S2 for 95 % confidence intervals.

= .001, but negatively with ATC, $r = -0.16$, 95 % CI [−0.28, −0.03], $p = .017$ (see Note 6 in supplementary materials).

The NARQ-Adm latent variable correlated positively with ATC, $r = 0.21$, 95 % CI [0.08, 0.33], $p = .002$, implying that higher levels of narcissistic admiration were associated with more favourable attitudes toward cryptocurrencies. By contrast, NARQ-Adm failed to correlate significantly with ATS, $r = 0.06$, [−0.06, 0.19], $p = .290$. The difference between the two correlations was significant, $\Delta r = 0.15$, 95 % CI [0.02, 0.27], $p = .028$. In comparison to NARQ-Adm, narcissistic rivalry (NARQ-Riv) failed to correlate significantly with either ATS or ATC (see Table 2). Finally, weekly earnings correlated relatively negligibly (< 0.15) with all of the variables of interest (not considered further).

3.3. Structural Equation Model (SEM)

Next, we tested a SEM predicting ATS and ATC with the following predictors (see Fig. 1): general intelligence (IQ*g*), financial literacy, NARQ-Adm and NARQ-Riv. The model yielded good model close-fit, $\chi^2(91) = 190.19$, $p < .001$, CFI = 0.962, TLI = 0.950, SRMR = 0.052, RMSEA = 0.054.

First, the NARQ-Adm latent variable was associated with a significant, positive beta-weight predicting ATC, $\beta = 0.32$, 95 % CI [0.15, 0.46], $p = .001$, whereas the NARQ-Riv latent variable was associated with a significant, negative beta-weight predicting ATC, $\beta = -0.24$, 95 % CI [−0.39, −0.08], $p = .002$. We note that the NARQ-Adm and NAQR-Riv beta-weights were both numerically larger than the corresponding NARQ-Adm ($r = 0.21$) and NARQ-Riv ($r = 0.06$) total latent effects, suggesting the observation of cooperative suppression. In comparison to ATC, NARQ-Adm ($\beta = 0.15$, 95 % CI [−0.01, 0.29], $p = .063$) and NARQ-Riv ($\beta = -0.12$, 95 % CI [−0.27, 0.06], $p = .178$) failed to predict ATS significantly.

Financial literacy was a significant, unique, positive predictor of ATS, $\beta = 0.40$, 95 % CI [0.28, 0.53], $p = .001$, controlling for the effects of general intelligence, whereas general intelligence was not a significant, unique predictor of ATS, $\beta = -0.12$, 95 % CI [−0.30, 0.07], $p = .206$. By contrast, general intelligence was a significant, unique, negative predictor of ATC, $\beta = -0.27$, 95 % CI [−0.44, −0.09], $p = .003$, whereas financial literacy was not, $\beta = 0.10$, 95 % CI [−0.04, 0.26], $p = .168$. Finally, the ATS and ATC latent variables were associated with multiple $R^2 = 0.149$ (95 % CI [0.076, 0.214], $p = .009$) and $R^2 = 0.105$ (95 % CI [0.029, 0.189], $p = .006$). Thus, approximately 15 % and 11 % of the ATS and ATC true score variance was accounted for in the model, respectively.

4. Discussion

We examined the associations between individual differences characteristics (i.e., narcissism, intelligence and financial literacy) and attitudes toward stocks and cryptocurrencies. We found that, although intercorrelated positively, narcissistic admiration and narcissistic rivalry differentially predicted attitudes toward cryptocurrencies (admiration, positively; rivalry, negatively), but both failed to associate significantly with attitudes toward stocks. Furthermore, financial literacy was a unique, positive predictor of attitudes toward stocks, whereas intelligence was a unique, negative predictor of attitudes toward cryptocurrencies. We discuss each of these results next.

4.1. Attitudes toward stocks and cryptocurrencies

We observed cryptocurrency and stock ownership rates of 12 % and 17 %, respectively, aligning with percentages reported in other countries with larger samples (Fujiki, 2021; Kim et al., 2019; Zhao & Zhang, 2021). Thus, approximately 1 in 10 young adults in Australia own cryptocurrencies: a substantial number, given cryptocurrency volatility (Shen et al., 2020).

Despite strong, opposing views in the financial community (Lou, 2022), attitudes toward stocks and cryptocurrencies intercorrelated appreciably ($r = 0.48$). The scatter plot suggested a necessary but not sufficient condition association. That is, those with a favourable attitude toward stocks tended to segment into two clusters (i.e., favourable versus non-favourable cryptocurrency attitudes), underscoring the divisiveness of cryptocurrencies (Lou, 2022). Political orientation might partially explain the nature of the scatter plot we observed in this investigation (Lou, 2022).

4.2. Narcissism and attitudes toward stocks and cryptocurrencies

Consistent with Martin et al. (2022), we found that narcissism associated with attitudes toward cryptocurrencies. We extend Martin et al. (2022) by showing that only narcissistic admiration, and not narcissistic rivalry, correlated positively with attitudes toward cryptocurrencies. Our results support theories that contend narcissistic admiration is characterised by hypersensitivities to reward (Grapsas et al., 2020), whereas narcissistic rivalry is characterised by a fear of failure orientation (Lange et al., 2016).

By comparison, neither narcissistic admiration nor narcissistic rivalry correlated significantly with attitudes toward stocks. Thus, we again extend Martin et al. (2022) by demonstrating discriminant validity with respect to narcissism as a predictor of attitudes toward investments. As cryptocurrencies are more volatile than stocks (Shen et al., 2020), those who score higher on narcissistic admiration may view such volatility as an especially attractive opportunity to gain wealth and prestige. Moreover, due to their novelty and complexity, cryptocurrency investors might be seen as intelligent, appealing to admiration-focused narcissists (Zajenkowski, Czarna, et al., 2020). Investments in stocks also provide opportunities to gain wealth, therefore, a positive effect between narcissistic admiration and attitudes toward stocks was expected. We acknowledge that narcissistic admiration yielded a positive, though non-significant, effect onto attitudes toward stocks ($\beta = 0.15$). Therefore, we do not feel confident contending the absence of an effect.

Like others (e.g., Gignac & Zajenkowski, 2021), we found narcissistic admiration correlated appreciatively with narcissistic rivalry ($r \approx 0.50$). When we estimated our SEM (i.e., unique effects), a cooperative suppression effect was observed, supporting further the approach/avoidance dispositional distinction between narcissistic admiration and rivalry (Grapsas et al., 2020; Lange et al., 2016). Specifically, the narcissistic rivalry effect onto attitudes toward cryptocurrencies was negative and statistically significant (i.e., −0.24; compare total effect 0.00), and the narcissistic admiration effect onto attitudes toward cryptocurrencies increased to 0.32 (total effect 0.21).

Together, narcissistic admiration and narcissistic rivalry accounted for 6 % of the variance in attitudes toward cryptocurrencies,[1] substantially more than the 1 % effect reported by Martin et al. (2022), thus underscoring the benefits of adopting a multidimensional (latent) approach to the measurement of narcissism. The positive association between narcissistic admiration and attitudes toward cryptocurrencies may prove disadvantageous, as narcissistic admiration is associated with ability overestimation (Zajenkowski & Gignac, 2021). Therefore, cryptocurrencies may attract a disproportionate number of (narcissistic) investors who are miscalculating the risks: a concerning possibility, as cryptocurrency exchanges are less regulated than other investment platforms (Afzal & Asif, 2019).

4.3. Financial literacy and attitudes toward stocks and cryptocurrencies

Consistent with research on financial literacy and stock ownership (Kim et al., 2019; Thomas & Spataro, 2018), we found financial literacy correlated positively with attitudes toward stocks. Thus, greater levels of financial literacy may facilitate a greater appreciation of stocks, assuming financial literacy → attitudes toward stocks. In contrast to stocks, financial literacy yielded a non-significant association with attitudes toward cryptocurrencies, in line with the existing (ownership) research that has yielded small and inconsistent effects (Fujiki, 2021; Zhao & Zhang, 2021). Thus, the effect between financial literacy and attitudes toward cryptocurrency may be near zero.

Similar to prior studies (e.g., Gignac et al., 2023; Lin & Bates, 2022), we observed a positive correlation between intelligence and financial literacy. Additionally, financial literacy yielded a factor loading on *g* comparable to other established intelligence indicators, suggesting its potential inclusion in the Cattell-Horn-Carroll intelligence model.

4.4. Intelligence and attitudes toward stocks and cryptocurrencies

Intelligence correlated positively with stock attitudes, consistent with the higher intelligence of stock owners compared to non-owners (Grinblatt et al., 2011). Our contribution is unique in that we examined attitudes, rather than simply ownership (impacted by opportunity). Thus, taken together, intelligence appears to be associated positively with attitudes toward investing in stocks, potentially (partly) explaining the positive association between intelligence and wealth accumulation (Banks et al., 2010). Precisely why relatively higher levels of intelligence facilitate a more positive attitude toward stocks remains to be determined. It may be a combination of factors, including better comprehension of risk and the capacity to calculate potential returns. Correspondingly, in our model, only financial literacy yielded a unique effect onto attitudes toward stocks, suggesting that financial literacy may mediate the effect between intelligence and attitudes toward stocks (see also Gignac et al., 2023).

In contrast to stocks, intelligence associated negatively with attitudes toward cryptocurrencies. Intelligence correlates positively with scepticism (Ståhl & Van Prooijen, 2018). As cryptocurrencies are a relatively new and unregulated asset class, it is possible that relatively intelligent people tend to view the asset class with a greater degree of scepticism. When we controlled for individual differences in financial literacy, the effect of intelligence onto attitudes toward cryptocurrencies increased, in comparison to the corresponding total effect (i.e., −0.16 to −0.27; a suppressor effect). We speculate that some factors that may dispose people to be interested in cryptocurrencies may also facilitate the acquisition of some financial literacy, but not a corresponding increase in intelligence. Thus, when controlling for financial literacy, the effect of intelligence revealed a more pure (non-suppressed) effect.

4.5. Limitations

As cryptocurrencies vary in design and purpose, our results may not generalise to attitudes to all types of cryptocurrencies. Our sample was comprised of university students, thus, it was not entirely representative of the population, and some range restriction in cognitive abilities may be expected. Furthermore, a more comprehensive approach to intelligence measurement (i.e., full test battery) may have yielded results different to those we reported. Additionally, conventional approaches to measuring financial literacy tend to tap principles that have a long-term horizon (e.g., impact of inflation on the value of money; compound growth, etc.), which may have reduced the chances of observing a positive association with attitudes toward cryptocurrencies (i.e., attractive for potential short-term gains). Also, participants completed the attitudes to cryptocurrency and stocks questionnaires last in the testing protocol, which may have affected the results to some degree (e.g., less valid responses due to respondent fatigue). Finally, elevated cryptocurrency and growth stock price fluctuations during the data collection period might have temporarily distorted some participant attitudes toward these investments.

[1] See Note 7 (supplementary materials).

5. Conclusion

Narcissistic admiration is positively and uniquely associated with attitudes toward cryptocurrencies - a potential concern, given the substantial volatility associated with cryptocurrencies, and the fact that narcissistic admiration is associated positively with the overestimation of abilities (Zajenkowski, Leniarska et al., 2020). Therefore, those who score high on narcissistic admiration may make unjustifiable assessments of the value of a cryptocurrency investment opportunity, particularly if they also score low on narcissistic rivalry and intelligence.

Declaration of generative AI and AI-assisted technologies in the writing process

Statement: During the preparation of this work Gilles Gignac used ChatGPT in order to generate suggestions to write some sentences more concisely. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Ethical statement

All procedures performed in studies involving human participants were in accordance with the ethical standards of the Ethics Committee from the institutions involved, and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Informed consent

Informed consent was obtained from all individual participants included in the study.

CRediT authorship contribution statement

Gilles E. Gignac: Conceptualization, Methodology, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Visualization, Project administration. **Chloe Jones:** Investigation, Writing – review & editing. **Natalie Mason:** Investigation. **Isabelle Yuen:** Investigation. **Marcin Zajenkowski:** Writing – review & editing.

Declaration of competing interest

On behalf of all authors, the corresponding author states that there is no conflict of interest. All authors have appropriate permissions and rights and also accept order of authorship.

Data availability

The data that support the findings of this study are openly available in the Open Science Framework (OSF) at https://osf.io/c6gq5/.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.paid.2023.112382.

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